

Detecting Whether a Document Is in an LLM's Training Data or Retrieval Index

Executive summary

There are known techniques for testing whether a document was used by, stored behind, or memorized by a large language model, but **the answer depends critically on what “indexed” means**. In practice, at least four distinct questions are often conflated:

1. **Pretraining membership:** Was some version of the document included in the corpus used to train the model's weights?
2. **Memorization/extractability:** Can the trained model reproduce distinctive portions of that document, possibly verbatim?
3. **Retrieval-index membership:** Is the document or one of its chunks currently stored in a RAG/ vector/keyword retrieval system used at inference time?
4. **Service retention:** Is a submitted document retained in logs, conversation state, caches, uploaded-file stores, or feedback systems?

These are not equivalent. A transformer's weights are not ordinarily a searchable “index” of its training documents; training turns statistical information about the corpus into parameters. By contrast, RAG systems use an actual external knowledge base or retrieval database whose documents or chunks are retrieved and added to model context. Service-side retention is yet another storage layer. Anthropic, for example, describes RAG as retrieving relevant material from a knowledge base and appending it to a prompt, while current OpenAI, Anthropic and Google API documentation separately distinguishes model training from logging and storage. ¹

The strongest conclusion from the literature is that **there is currently no universally reliable, black-box post-hoc test that can prove an arbitrary document was in a proprietary LLM's pretraining set**. Membership-inference attacks can provide evidence, sometimes quite useful evidence, but their false-positive rate is difficult to establish against the correct counterfactual model—namely, the same model trained the same way without the document. Zhang, Das, Kamath and Tramèr argue that this makes generic post-hoc membership inference unsound as a formal “training-data proof.” They identify prospective canaries/watermarks and actual extraction of substantial distinctive text as much stronger forms of evidence. ²

The practical evidence hierarchy is roughly:

Evidence	What it establishes	Evidentiary strength
Model knows facts from the document	Almost nothing about membership; facts may come from elsewhere	Very weak

Evidence	What it establishes	Evidentiary strength
Lower loss / Min-K score than controls	Behavior statistically consistent with membership	Weak-moderate
Multiple distinctive prefixes produce unusually accurate continuations	Evidence of memorization/ extractability	Moderate-strong
Long, uncommon passages reproduced verbatim and independently verified	Strong evidence of memorization, often compelling evidence of training exposure	Strong
Pre-registered random canary/ watermark detected with calibrated significance	Statistical evidence designed specifically for provenance	Very strong under its assumptions
RAG returns a unique document/ chunk or trusted source ID	Document is in, or accessible to, the retrieval system	Strong-definitive
Internal corpus/index manifest, content hash, ingestion log	Direct membership evidence	Definitive for that version of the system

The most practically useful black-box test for an ordinary document is therefore **not simply “ask the model whether it has seen this document.”** Instead, use many independent, distinctive passages, vary prefix length, request short deterministic continuations, compare against matched negative controls, and—where token log probabilities are available—combine this with Min-K/likelihood or neighborhood-comparison scores. Carlini et al. found extractability increases with model size, duplicate count and prefix/context length; Shi et al.’s Min-K% method exploits the tendency of nonmembers to contain a few unusually low-probability tokens. ³

For **RAG/index membership**, the situation is considerably more favorable to an auditor. Research has demonstrated both black- and gray-box membership inference against retrieval databases, and newer attacks formulate several natural questions about distinctive facts in a target document and infer membership from the system’s ability to answer them. Anderson, Amit and Goldsteen demonstrated black- and gray-box RAG membership inference across multiple generators; subsequent work has pushed toward natural, query-efficient attacks. ⁴

A critical practical point is that **setting `max_tokens` or another output-token limit does not itself reveal training data**. The scientifically useful variables are instead *prefix length*, *suffix length*, decoding method, repetition across passages, and comparison against controls. Longer, sufficiently distinctive prefixes can make an already-memorized continuation easier to elicit, whereas simply increasing the output budget mainly gives the model more opportunity to generate text. ⁵

flowchart TD

```

A[What does "present" mean?] --> B[Pretraining corpus]
A --> C[RAG / retrieval index]
A --> D[Logs, files, cache or conversation state]

```

```

B --> E{Access}
E -->|Output only| F[Prefix continuation / extraction probes]
E -->|Token probabilities| G[Loss, Min-K, likelihood-ratio tests]
E -->|Weights| H[Exact token scoring, Min-K++, canary/watermark analysis]
E -->|Training pipeline| I[Corpus manifests, hashes, lineage]

C --> J{Access}
J -->|Output only| K[Distinctive-fact / entailment MIA]
J -->|Retriever metadata| L[Inspect retrieved chunks and source IDs]
J -->|Index owner| M[Direct hash / document-ID lookup]

D --> N[Inspect provider retention controls and audit logs]

```

This separation reflects the threat models studied in training-data extraction, membership inference and RAG-membership work. ⁶

What “indexed” means and the relevant threat models

Pretraining membership, memorization and extraction

Let a candidate document be D . **Membership** asks whether D , a chunk of D , or a sufficiently close transformed copy occurred in the training corpus. That definition is already less clean than it sounds because pipelines can normalize markup, deduplicate near-copies, filter sections, concatenate documents, tokenize them differently, or retain only pieces of a source. Meta, for example, documents heuristic filtering, semantic deduplication and quality classification in the Llama 3 training pipeline. ⁷

Memorization is a behavioral property of the trained model and does not follow automatically from membership. Carlini et al. operationalize *extractable memorization* by asking whether training text can be reproduced when the model receives an appropriate prefix. Their experiments found memorization grows with model capacity, duplication and amount of prefix context. ⁸

This distinction matters because:

training membership $\nRightarrow$ black-box extractability

and, conversely, a familiar-looking answer does not uniquely identify the particular source from which it was learned. Common text may appear in many sources. Kandpal, Wallace and Raffel found duplication to be an especially strong driver: in their experiments, a sequence appearing ten times was generated roughly $1000\times$ as frequently as one appearing once, while membership-detection methods approached chance performance on nonduplicated sequences. ⁹

Retrieval/index membership

For a RAG system, “indexed” has a much more literal meaning. A document may be parsed into chunks, embedded or keyword-indexed and stored in a retrieval database. A query retrieves some of those chunks and supplies them as model context. ¹⁰

Here the hypothesis is typically

$$H_0 : D \notin R, \quad H_1 : D \in R$$

where R is the retrieval corpus. Anderson et al. showed that the final answers of a RAG system can leak this membership even without direct access to the database. Their experiments cover both **black-box** and **gray-box** settings. ¹¹

For an owner/operator of the retrieval system, inference should normally be unnecessary: inspecting ingestion records, canonical document IDs, chunk text or content hashes is far stronger than asking the LLM.

Retention is not training

A third category is operational storage. Current provider documentation illustrates why this must be kept separate from training-data membership.

As of September 8, 2026, OpenAI states that data submitted through its API is not used to train or improve models by default unless a customer explicitly opts in. It separately states that default abuse-monitoring logs may include prompts and responses and may be retained for up to 30 days, subject to stated exceptions. ¹²

Anthropic states that inputs and outputs from commercial products such as the Anthropic API and Claude for Work are not used to train its models by default. Explicitly submitted feedback is treated differently and can be stored for substantially longer periods. ¹³

Google states that prompts, associated instructions, cached content, files and responses from Gemini Paid Services are not used to improve its products; its documentation separately describes abuse logging, state storage, File API storage and special retention associated with Search/Maps grounding. ¹⁴

Thus:

retained / indexed for RAG / used for training. =

Access levels

The useful access taxonomy is:

Output-only black box gives you generated text but not token probabilities. Prefix continuation, extraction attempts, RAG question-based inference and some contamination tests are possible. ¹⁵

API black box with log probabilities permits loss/perplexity, Min-K and calibrated likelihood tests. This is substantially more informative than output text alone. The original membership-inference literature already treats confidence/probability access as an important black-box signal. ¹⁶

Open weights let an auditor compute exact next-token probabilities and arbitrary statistics such as Min-K+⁺, canary rank or watermark residuals without relying on a vendor's API surface. Open weights still do **not** provide a list of the training documents. ¹⁷

Training-pipeline or retrieval-index access is qualitatively stronger. Dataset manifests, hashes, ingestion logs, checkpoints and database records can directly answer membership questions rather than infer them from model behavior.

Comparison of techniques

technique	access required	steps	effectiveness	limitations	references
Token-limited prefix continuation	Output-only API	Select distinctive prefix → set deterministic decoding and a fixed short output limit → ask for continuation → compare with true suffix → repeat across passages/ prefix lengths	Exact reproduction of several rare continuations is strong evidence of extractable memorization; longer contexts can increase extraction	High false-negative rate; common quotations create false positives; safety/ copyright filters can suppress true memorization; output-token count itself is not a detector	Carlini et al. on extractability and context length. ⁵
Raw loss/ perplexity threshold	Token probabilities or weights	Compute $-\log p(D)$ or token-average NLL → classify unusually low-loss documents as members	Historically useful baseline	Intrinsically easy/predictable text also has low loss, producing high FPR; document length/domain strongly confound the score	Mattern et al. explicitly identify high FPR from uncalibrated model scores. ¹⁸

technique	access required	steps	effectiveness	limitations	references
Min-K% probability	Per-token log probabilities or weights	Compute each token's conditional probability → take the lowest-probability $k\%$ → average them → compare with controls/ threshold	Shi et al. reported a 7.4% improvement over prior methods on WikiMIA and applied the method to books, benchmark contamination and unlearning audits	Still a post-hoc MIA; calibration and distribution shift remain serious issues; proprietary APIs may not expose logprobs	Shi et al. ¹⁹
Min-K%++ / calibrated token statistics	Rich token-probability access, usually open weights	Calculate target-token scores relative to the model's overall predictive distribution, then aggregate unusually informative tokens	Improves over vanilla Min-K in reported pretraining-detection experiments	Same fundamental membership-proof problem; requires richer probability information	Min-K%++. ²⁰
Reference-model likelihood ratio	Token probabilities plus a suitable reference model	Compute target-model likelihood → compute likelihood under comparable reference/ nonmember model → use the difference/ ratio as membership statistic	Can remove some intrinsic text-difficulty effects and outperform raw loss	Strong assumption that reference model/ distribution is appropriate; attacks become fragile under reference-distribution mismatch	Ye et al.; Mattern et al. ²¹

technique	access required	steps	effectiveness	limitations	references
Neighborhood comparison	Token probabilities; ability to generate perturbed neighbors	Generate meaning-preserving/ syntactic neighbors of target → score target and neighbors → test whether target is anomalously easy for suspect model	Reported competitive with strong reference-model attacks without requiring access to original training distribution	Quality of synthetic neighbors matters; still cannot establish the exact counterfactual FPR for a proprietary model	Mattern et al. ¹⁸
Canary / exposure test	Prospective control of candidate data; probability access is ideal	Before training/ publication, randomly choose a secret from a prespecified space → insert it → later rank the planted secret among alternatives under the suspect model	Designed for statistically calibrated detection of rare memorization; much stronger evidence than arbitrary post-hoc MIA	Must be planted beforehand; insertion frequency and random space must be recorded; preprocessing may remove it	<i>The Secret Sharer</i> ; Zhang et al. ²²

technique	access required	steps	effectiveness	limitations	references
Targeted extraction / verbatim-regurgitation test	Output API; stronger with logprobs/weights	Supply known distinctive prefixes or generate candidate outputs → identify unusual low-perplexity candidates → verify against the source/corpus	Long, distinctive, verified verbatim extraction is among the strongest post-hoc signals. GPT-2 experiments recovered hundreds of training examples; later work showed substantially larger-scale extraction	Potentially expensive; provider safeguards/rate limits interfere; non-extraction does not show nonmembership; ethical/copyright/privacy risk	Carlini et al.; Nasr et al. ²³
RAG membership probing	Output-only or gray-box RAG endpoint	Build questions from document-specific details → query naturally → score whether answers reveal those details or explicitly acknowledge context membership → aggregate multiple probes	Demonstrated across multiple generators and datasets in both black- and gray-box settings	Parametric knowledge can masquerade as retrieval; retriever recall can cause false negatives; browsing or other data sources must be isolated	Anderson et al.; newer E-MIA/MENTA work. ⁴

technique	access required	steps	effectiveness	limitations	references
Prefix/suffix semantic RAG MIA	Output-only RAG	Give part of target text or a question derived from it → measure similarity/ entailment between answer and held-out target content	Can detect retrieved content without requiring raw retriever scores	Semantic overlap can arise from general model knowledge; effectiveness is system-specific	S ² MIA and related RAG attacks. ²⁴
Canonical-order / permutation contamination test	Usually sequence-likelihood access	Randomly permute benchmark examples → compare likelihood/rank of canonical ordering against permutations → test exchangeability	Provides a statistically motivated test for benchmark-level contamination under its assumptions	Tests a structured dataset/order; not an arbitrary standalone document; requires a canonical sequence and appropriate scoring access	Oren et al., ICLR 2024. ²⁵
Watermarked training-data “radioactivity”	Prospective watermarking; strongest with open weights	Generate/ release watermarked text → later test suspect model for residual watermark statistics	Sander et al. obtained $p < 10^{-5}$ with an open-weight suspect model even when only 5% of training text was watermarked in one studied setting	Prospective only; results depend on watermark robustness, amount of marked data and training procedure; ordinary output watermarks do not retroactively identify arbitrary documents	Sander et al. ²⁶

technique	access required	steps	effectiveness	limitations	references
Direct index/ corpus audit	Retrieval DB, ingestion system or training-data lineage access	Normalize candidate as ingestion does → compare canonical hash/ID/text/ chunks → inspect versioned ingestion/ deletion records	Potentially definitive	Available only to system owner/ provider; need to account for near- duplicates and historical versions	This is an architectural/ data-lineage check rather than an inference attack; RAG databases are separate from parametric memory. 27

Interpreting effectiveness correctly

For MIA evaluation, **ROC-AUC alone is not enough**. A useful audit should report at least TPR/recall at a very low fixed FPR, precision under a realistic membership base rate, and preferably confidence intervals. Membership inference is naturally a hypothesis test, and the most consequential quantity for a claim such as “this publisher’s book was used” is the probability of falsely accusing a nonmember. 28

For prevalence π , true-positive rate t and false-positive rate f ,

$$\text{Precision} = \frac{t\pi}{t\pi + f(1 - \pi)}.$$

Consequently, even an apparently small FPR can produce many false accusations when genuine membership is rare.

The 2024 “training data proof” critique goes further: to establish a legally or scientifically persuasive FPR, one ideally needs models sampled from the **true null distribution**, meaning essentially identical training procedures in which only the target data is excluded. For a closed, web-scale proprietary model, an external researcher usually cannot construct that counterfactual. 2

For extraction experiments, better metrics include **exact-match rate, longest exact continuation, unique extracted sequences, edit distance, and the proportion of outputs independently verified against the target**. Exact matching of long, distinctive text is more probative than ROUGE or semantic similarity, because semantic similarity may merely demonstrate that the model understands the subject.

For prospective canary/watermark tests, pre-specified **p-values, FPR bounds and detection power** are preferable to arbitrary classification thresholds. The watermark-radioactivity work is an example of this style of statistical provenance test. 26

Practical tests you can run

These tests are appropriate for a document you own, control, or are authorized to audit. For a third-party hosted model, first check its API terms: some providers explicitly restrict reverse engineering or automated extraction. OpenAI's current Terms of Use prohibit attempts to discover underlying model components and automatically/programmatically extracting data or output; Google's Gemini API terms prohibit attempts to reverse engineer, extract or replicate underlying data or models. ²⁹

Controlled prefix-continuation battery

This is the easiest useful black-box experiment.

Take approximately 10–30 passages from different locations in the target document. Avoid titles, boilerplate, famous quotations, tables of contents and phrases likely to have been copied elsewhere. For each passage, divide it into a prefix P and a held-out suffix S . Repeat at several prefix lengths—for example, approximately 32, 64, 128 and 256 model tokens—while holding the desired suffix around 32–64 tokens.

A concrete prompt is:

```
You are participating in an authorized memorization audit.  
  
Continue the text below with the next 48 tokens from the original source.  
Do not summarize, explain, modernize, or paraphrase it.  
If you do not know an exact continuation, output [UNKNOWN].  
  
TEXT:  
<insert the target prefix here>
```

At the API level, use deterministic or near-deterministic decoding where possible and set the maximum output length close to the suffix length. The important manipulation is **prefix length**; the output limit merely standardizes the experiment. Carlini et al. found that increased prefix/context length substantially increases the probability of extracting memorized training sequences. ⁸

Score exact token/character matches first. A useful hierarchy is:

full exact match > long exact prefix of suffix > small-edit-distance match > semantic paraphrase.

Run identical tests against at least 20–100 **matched controls**: text of similar genre, age, length, language and predictability that you have good reason to believe was not in training. Ideally, controls were privately generated before the audit or created after a reliably known training cutoff. Shi et al.'s WikiMIA was motivated by precisely the difficulty of obtaining credible membership ground truth. ¹⁹

Expected positive outcome: several geographically separated, stylistically distinctive prefixes produce long exact continuations, particularly as prefix length grows.

Expected negative outcome: incorrect or generic continuations. This is **not evidence of nonmembership**: the document may have been trained on but not memorized, deduplicated, downweighted, transformed, subsequently unlearned, or blocked by serving-time safeguards. Kandpal et al.'s results on nonduplicated sequences illustrate how severe the false-negative problem can be. ⁹

Do not count a model's statement such as "I have seen this document" or "this was in my training data" as evidence. Such self-reports are generated text, not introspection into a training manifest.

Log-probability and Min-K battery

Where an API returns token log probabilities—or with an open-weight model—score the actual candidate document rather than asking the model to generate it.

For tokens $x_1, \dots, x_n$,

$$\ell_i = \log p(x_i \mid x_1, \dots, x_{i-1}).$$

Raw mean log likelihood is

$$L(D) = \frac{1}{n} \sum_i \ell_i.$$

Members often have higher likelihood, but this statistic is heavily confounded by intrinsic text difficulty. Mattern et al. found simple score thresholding can have high false-positive rates for precisely this reason. ¹⁸

For Min-K%, sort the ℓ_i values and average only the least likely $k\%$:

$$M_k(D) = \text{mean}(\text{lowest } k\% \{\ell_1, \dots, \ell_n\}).$$

Shi et al.'s intuition is that an unseen sequence is more likely to contain a handful of surprisingly low-probability tokens; a training example has less pronounced "surprise" in this tail. ¹⁹

In practice, calculate $k = 10, 20, 30\%$ rather than selecting one value after seeing the result. Compute the statistic separately on multiple non-overlapping chunks and compare every statistic with your matched negative controls.

A practical decision rule might be:

Do not call membership because one passage scores well.

Flag the document for stronger investigation only if:

- most independent chunks score in the extreme high tail of matched controls,
- the result is stable across several values of k ,

- prefix extraction independently shows unusually exact continuations, and
- there is no obvious source/domain/age mismatch between target and controls.

That is intentionally conservative because Min-K is a **detector**, not a definitive provenance certificate. 30

Reference and neighborhood likelihood-ratio tests

Where you can run a second model, calculate something approximately equivalent to

$$S(D) = \log p_{\text{suspect}}(D) - \log p_{\text{reference}}(D).$$

The hypothesis is not merely that D is predictable, but that it is **disproportionately predictable to the suspect model** relative to a model that learned similar general language patterns.

Alternatively, generate multiple close neighbors $D'_1, \dots, D'_m$ through minor paraphrasing or substitutions and compare

$$S_{\text{neigh}}(D) = L(D) - \frac{1}{m} \sum_j L(D'_j).$$

Mattern et al.'s neighborhood attack was designed specifically to avoid the need to possess a reference training distribution and outperformed several reference-free baselines in their experiments. 18

The central risk is calibration: a suspect model and reference model may differ in tokenizer, model family, training era or specialization. A large likelihood ratio can therefore represent architecture/domain differences rather than membership. This is one reason not to interpret such tests as proof. 31

RAG/index membership test

If your real concern is whether a private document is in a retrieval system, use an entirely different experiment.

First identify approximately 5–15 **hard-to-guess details** from the document: unusual proper nouns, artificial identifiers, exact numerical relationships, project-specific terminology, or facts whose combination is unique. Avoid facts easily available on the public web.

Then ask natural questions independently:

Using only the connected knowledge base, what identifier is associated with the "Orchid reconciliation" procedure?

Give the answer and the source title or source identifier used.

Using only the connected knowledge base, which three conditions must hold before the "Orchid reconciliation" procedure is approved?

Cite the source retrieved for your answer.

According to the connected documents, what relationship is stated between Project Orchid, batch K7VQ, and the May reconciliation event?

Aggregate the answers rather than relying on one. This resembles the direction of recent document-level RAG MIAs, which transform distinctive document evidence into sets of questions and then infer membership from answer quality or entailment. ³²

The experimental controls are crucial. Disable ordinary web browsing, if possible. Use a new/private document so that general pretraining cannot plausibly contain it. Ask matched questions about similar documents known **not** to have been ingested. If the platform supplies trusted, system-generated citations or retrieved chunk IDs, preserve those separately from LLM-generated citation text.

A strong positive is a unique answer accompanied by a trustworthy retriever-generated source ID pointing to the target. A weak positive is merely that the model knows the answer. Anderson et al.'s work demonstrates that RAG membership can leak through final generated responses even in black-box conditions, but it also means RAG membership attacks must separate retrieved information from the base model's parametric knowledge. ¹¹

Prospective canary test

For documents that have **not yet been published or ingested**, a canary is much stronger than trying to audit an arbitrary historical document afterward.

Generate a uniformly random value from a recorded format, for example:

Audit nonce for Cedar release:
K7VQ-29PF-4RXT-8HNC

Place it naturally in the controlled document, record the candidate universe and insertion count, timestamp the design, and ideally commit cryptographically to the selected canary before the document is released.

After the suspected training/ingestion event, test the probability or rank of the real canary against alternatives drawn from exactly the same random space. The classic “Secret Sharer” work introduced the **exposure** framework for this type of audit, and the later training-data-proof analysis identifies randomized canaries as one of the few ways to obtain a sounder statistical provenance argument. ²²

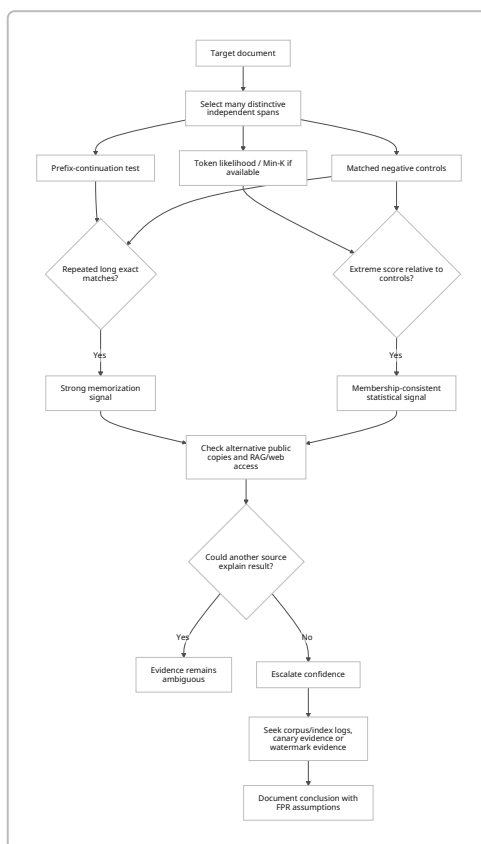
Conceptually,

$$\text{exposure} \approx \log_2 |\mathcal{R}| - \log_2(\text{rank}(\text{canary})),$$

where $\mathcal{R}$ is the prespecified random candidate space. A canary ranked vastly higher than random alternatives is much more informative than an arbitrary natural-language passage whose baseline predictability was unknown.

Combined protocol

For a serious investigation, use triangulation rather than selecting whichever detector gives the desired answer.



The key methodological principle is that **independent forms of evidence should agree**. Exact extraction, low-tail likelihood anomalies and retrieval metadata are substantially more persuasive together than any one score in isolation. The literature's strongest criticism of post-hoc MIAs is precisely that impressive benchmark accuracy does not automatically translate into a defensible false-positive probability for one real-world proprietary model. ²

Defenses and mitigations

Training-time controls

Deduplication is one of the most empirically established defenses against verbatim extraction.

Kandpal et al. found extraction susceptibility grows superlinearly with duplicate count and that detection became near-chance on nonduplicated sequences in their experiments; separate work on deduplicated language-model datasets likewise found sharply reduced memorized output. ³³

Semantic as well as exact deduplication is preferable because the same material often appears with formatting changes or mirrors. Meta publicly describes semantic deduplication as part of the Llama 3 pretraining-data pipeline. ⁷

Differential privacy is the strongest formal mechanism in this list. Rather than merely trying to hide memorization at inference, DP training bounds how much any individual example can influence the released model. Google Research/DeepMind's VaultGemma work demonstrates full differentially private pretraining of a 1B-parameter language model and explicitly frames the approach as protection against memorization and training-data extraction. DP fine-tuning alone cannot retroactively protect data already memorized during nonprivate pretraining. ³⁴

The tradeoff is that strong privacy guarantees impose optimization, compute and utility constraints; consequently, DP should be treated as a model-development architecture decision rather than a post-deployment filter. ³⁴

Sensitive-data filtering—PII detection, copyright/licensing controls, secrets scanning and quality filters—reduces the amount of problematic material available to memorize but provides no formal guarantee that every undesirable example is removed.

Serving-time controls

Output limits, rate limits, suppression of anomalously long verbatim continuations, copyright/PII classifiers and limiting high-resolution token-probability APIs can make extraction harder. They are mitigations against the **attack channel**, however, not evidence that the information was absent from training. Carlini et al.'s and Nasr et al.'s extraction results demonstrate the distinction between latent memorization and the methods needed to elicit it. ²³

Context-window limits are a weak privacy defense. Because longer prefixes can improve targeted extractability, restricting arbitrarily large attacker-controlled prefixes may remove one useful attack lever, but a short distinctive prefix can still trigger memorization. Context limits should therefore not substitute for deduplication or DP. ⁸

Likewise, withholding log probabilities reduces the effectiveness of Min-K and likelihood-ratio audits but does not prevent an output-only extraction attack.

RAG defenses

For RAG, **authorization before retrieval** is much more important than prompt wording. Enforce tenant isolation and document-level ACLs in the retrieval layer, so an unauthorized document never enters the model context.

Other useful measures include restricting the retriever's query scope to authenticated collections, enforcing authorization again after retrieval, treating retrieved content as untrusted data rather than instructions, limiting unnecessary raw chunk exposure, monitoring repeated probes about obscure entities and maintaining a distinction between system-generated provenance metadata and LLM-generated citations.

Anderson et al. found that adding defensive instructions to the RAG template could be effective for some dataset/model combinations, but prompt instructions are inherently a softer protection than retrieval-layer access control. ¹¹

Provenance controls

For organizations that need to prove later whether their intellectual property was used, **prospective controls are much better than forensic reconstruction**.

Use versioned document hashes, signed publication timestamps, unique document identifiers, ingestion manifests, pre-registered random canaries and, where appropriate, statistically testable data watermarks. The “radioactivity” work is particularly important because it shows that a watermark in training text can leave a statistically detectable residual in a later model; in its reported open-weight experiment, a signal with $p < 10^{-5}$ remained when watermarked text comprised only 5% of training material. ²⁶

Watermarking should not be confused with output watermarking. An ordinary watermark intended to identify AI-generated output does not automatically tell you whether an arbitrary preexisting document entered a training set. The provenance scheme must be designed **before the suspect training event**. ³⁵

Operational data handling

Provider-side training opt-outs and zero-data-retention controls address a different risk but are still relevant if the concern is that confidential documents submitted to a model may later become training material.

As currently documented, OpenAI's API, Anthropic's commercial products and Google's Gemini Paid Services all state forms of default separation between customer API content and general model training/product improvement, although the detailed retention, feedback, grounding and state-storage rules differ materially. ³⁶

For sensitive deployments, therefore, preserve the provider's terms and data-control documentation that was operative **at the date of submission**; today's policy does not itself establish how a historical document was handled.

Legal, ethical, and incident context

Evidentiary and legal limits

A membership score should not be presented as a factual accusation that a vendor copied a document unless the underlying false-positive assumptions are defensible. The 2024 Zhang et al. analysis was written partly in response to attempts to use MIAs as evidence in disputes around foundation-model training and concludes that arbitrary post-hoc MIAs cannot in general supply the necessary counterfactual FPR. ²

The strongest defensible evidence is accordingly one of:

- direct records showing corpus/index membership;
- a prospectively randomized canary or watermark with a pre-specified hypothesis test; or

- sufficiently distinctive, long-form extraction for which a non-training explanation is implausible. ³⁷

That last category still requires care. A model could have encountered a mirror, quotation, pirated copy, cached version, dataset compilation or derivative containing the same passage, so extraction may establish **exposure to the text** more readily than the exact upstream provenance route.

Testing third-party services can also implicate contractual restrictions. OpenAI's current consumer terms restrict reverse engineering, automated/programmatic extraction, circumvention of rate limits and bypassing protective measures; Google's current Gemini API terms similarly prohibit attempting to extract or replicate underlying data or models. ²⁹

The safest forensic protocol therefore uses documents you own or are authorized to test, minimizes collection of unrelated output, avoids intentionally extracting other users' PII or proprietary text, respects service rate limits and preserves only the evidence necessary for the audit.

The “Deep research traces hacked” question

I did not find a recognized incident carrying the exact name “**Deep research traces hacked.**” The phrase appears most plausibly to conflate several quite different AI-security incidents.

The closest direct match involving a **Deep Research** product is **ShadowLeak**, disclosed by Radware on September 18, 2025. Researchers reported that an indirect prompt injection hidden in an email could influence ChatGPT's Deep Research agent when it had Gmail and browsing access and cause service-side exfiltration of sensitive mailbox material. Radware described the proof of concept as zero-click from the victim's perspective and reported a 100% success rate in its experimental setup. This was a **connector/agent prompt-injection and exfiltration vulnerability**, not proof that Gmail documents were incorporated into model training. ³⁸

A second possible source of confusion is the **DeepSeek database exposure** found by Wiz Research on January 29, 2025. Wiz reported an unauthenticated ClickHouse database containing more than a million log lines, including chat history, secret keys and backend information, with broad database access. Again, this was an exposed operational database/logging incident—not a demonstration that the exposed chats were encoded into DeepSeek model weights. ³⁹

A historically more relevant incident for **training-data memorization** is the sequence of language-model extraction studies. Carlini et al. demonstrated extraction of hundreds of verbatim GPT-2 training examples, including uncommon and sensitive-looking material; Nasr et al. later demonstrated much larger-scale extraction against several classes of model and reported a 2023 “divergence” behavior in ChatGPT that increased the rate of training-data-like regurgitation dramatically in their experiments. These incidents establish that serving-time safeguards do not necessarily imply absence of underlying memorization. ⁴⁰

These three cases illustrate why the storage layer must always be identified before drawing conclusions:

agent exfiltration /

Bottom-line forensic recommendation

For an **existing arbitrary document against a closed API**, I would treat the following as the highest-value sequence:

First, disable browsing/RAG/tools and run multi-span deterministic prefix continuations. **Second**, obtain token log probabilities if legitimately available and run Min-K plus a neighborhood or carefully matched reference analysis. **Third**, repeat with matched negative documents and report TPR/FPR-oriented statistics rather than a single membership score. **Fourth**, separately test any RAG layer using unique-fact queries and trusted retrieval metadata. **Fifth**, search for alternative public copies that could explain verbatim reproduction. Only repeated, long, distinctive extraction should materially raise confidence beyond “statistically consistent with membership.” ⁴¹

For **future documents**, pre-register a random canary or provenance watermark before release. This changes the problem from weak post-hoc inference into a deliberately designed statistical experiment and is the clearest direction in the literature for defensible proof. ⁴²

For a **RAG system you operate**, do not infer membership through the model at all: maintain canonical document hashes, ingestion/deletion logs and versioned chunk manifests and query those directly. Behavioral RAG MIA should be reserved for external auditing or validation that access-control boundaries are not leaking membership.

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